MINGYUE GUO

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EDUCATION

Texas A&M University, Texas, U.S. January 2023 - May 2027
Ph.D. Candidate in Mechanical Engineering *GPA 4.0/4.0*

- **Dissertation**: Developing practical learning-based controls for building energy systems
- **Advisor**: Dr. Zheng O'Neill, PhD, PE, FASHRAE, FIBPSA

Tongji University, Shanghai, China September 2019 - March 2022
M.S. in HVAC & Gas Engineering *overall GPA 4.37/5*

- **Thesis**: A hybrid energy consumption prediction model for office buildings based on heterogeneous data **(Excellent Master's Thesis Award)**
- **Advisor**: Peng Xu, PhD, PE, MBA, FIBPSA

Chongqing University, Chongqing, China September 2015 - June 2019
B.E. in Built Environment; Minor in Business Administration *GPA 3.69/4; rank: 2/113*

- **Thesis**: Preliminary Study on the Automatic Design of HVAC Systems Based on BIM
- **Supervisor**: Nan Li, Peng Xu

PROFESSIONAL EXPERIENCE

Research Assistant **Texas A&M University** 2023 - present
Mentor: Zheng O'Neill
Research included: occupancy-centric control field testing and results analysis, large-scale building energy simulation and data analysis, and intelligent controllers (e.g., MPC, RL, and LLM) for efficient energy management.

Machine Learning Research Intern **Hewlett Packard Enterprise** May - Aug. 2025
Mentors: Soumyendu Sarkar, Avisek Naug
Developed reinforcement learning control for data center liquid cooling systems.

Graduate Research Intern **Oak Ridge National Laboratory** Jun. - Aug. 2024
Mentors: Sen Huang, Jamie (Jianming) Lian
Developed control-oriented models and conducted field testing for model-predictive control strategies to optimize energy costs in a grid-interactive commercial building while maintaining thermal comfort.

Research Assistant **Hong Kong University of Science and Technology** Sep. - Dec. 2022
Conducted research on model predictive control to optimize the HVAC system of a commercial building.

Research Assistant **Tongji University** Sep. 2019 - Mar. 2022
Mentor: Peng Xu
Developed a hybrid energy prediction model using data-driven methods.
Conducted research on automatic BIM to BEM and HVAC system sizing.

Assistant Designer **CMCU Engineering Co. Ltd. Chongqing, China** *Jul. - Aug. 2017*
Assisted in the design of the HVAC system for a high-rise residential building in Guizhou, China.

SELECTED RESEARCH PROJECTS

Texas grid impact of replacing existing residential AC with heat pumps Jun. - Nov. 2026
Funded by Daikin, lead graduate researcher at TAMU

• Simulate the Texas grid impacts of replacing existing air-conditioning systems with federally compliant minimum-efficiency heat pumps and inverter-driven heat pumps using ResStock and EnergyPlus.

An Integrated Approach to Modeling, Decision-Making and Control for Energy Efficient Manufacturing Sep. 2023 - present

Funded by National Science Foundation (NSF), lead graduate researcher at TAMU

• Developed and validated control-oriented models, and implemented decision-making and adaptive controls for the HVAC systems of semiconductor cleanrooms.

Large-Scale Smart Meter Data Analysis May 2024 - present

Funded by Texas A&M University

• Data analysis for smart meter data from over 2 million consumers in the Greater Houston Area, Texas.
• Analyzed energy resilience in a changing climate through 2090 by developing data-driven models.
• Correlated electricity consumption with demographic factors and tree canopy.

Quantification of HVAC Energy Savings for Occupancy Sensing in Buildings Through an Innovative Testing Methodology Jan. 2023 - Aug. 2024

Funded by Advanced Research Projects Agency-Energy (ARPA-E), research assistant at TAMU

• Maintained and troubleshot field-testing systems in a single-family home and a commercial building.
• Analyzed the sensor accuracy, energy savings, and thermal comfort in three field tests.
• Extracted typical occupancy schedules from field-measured data using the K-means clustering algorithm.

Development and Validation of Smart Building Technology Modules for Academic and Professional Education Jan. 2023 - Dec. 2025

Funded by U.S. Department of Energy (DOE), research assistant at TAMU

• Polished and proofread the educational modules

Target-controlled Feedforward Management Technology for Green Buildings 2019 - 2021

Funded by the "13th Five-Year Plan" Key Research Program, China; research assistant in Tongji Univ.

• Developed an automatic tool to convert BIM (Building Information Modeling) in Revit to BEM (Building Energy Modeling) in EnergyPlus through an intermediate gbXML file.
• Enabled automatic gbXML check and IDF enrichment to ensure the success of the BIM to BEM conversion.

Automatic HVAC design 2020 - 2021

Funded by Tongji University Architectural Design Institute Co., Ltd.; research assistant in Tongji Univ.

• Developed an automatic configuration tool for the AHU, FCU, and chiller sizing in Python

JOURNAL PUBLICATIONS

Equal Contribution.

[1] **M. Guo**, Z. O'Neill, Z. Yang, J. Wen, W. Sun, Cleanroom environmental control systems: review of research landscape and control strategies, Energy and Buildings 365 (2026) 117639. `https://doi.org/10.1016/j.enbuild.2026.117639`.

[2] Z. Pang#, **M. Guo**#, Z. O'Neill, B. Smith-Cortez, Z. Yang, B. Dong, A longitudinal field study of sensor-driven occupancy-centric HVAC controls in an office building, Energy and Buildings 351 (2026) 116693. `https://doi.org/10.1016/j.enbuild.2025.116693`.

[3] **M. Guo**, Y. Choi, S.-M. Cheong, Z. O'Neill, Current and Future Residential Electricity Demand Using Large-Scale Smart Meter Data in a Changing Climate, Sustainable Cities and Society (2025) 106623. https://doi.org/10.1016/j.scs.2025.106623.

[4] **M. Guo**, B. Liu, S. Huang, J. Dong, J. Lian, Z. O'Neill, Deploying advanced supervisory control strategies to small- and medium-sized commercial buildings: Case study and lessons learned, Energy and Buildings (2025) 115710. https://doi.org/10.1016/j.enbuild.2025.115710.

[5] Z. Pang, **M. Guo**, Z. O'Neill, B. Smith-Cortez, Z. Yang, M. Liu, B. Dong, Long-Term field testing of the accuracy and HVAC energy savings potential of occupancy presence sensors in a single-family home, Energy and Buildings 328 (2025) 115161. https://doi.org/10.1016/j.enbuild.2024.115161.

[6] **M. Guo**, B. Smith-Cortez, Z. O'Neill, T. Firsich, Z. Yang, Exploring and field-demonstrating geofence-based occupancy-centric control in residential buildings, Energy and Buildings 320 (2024) 114595. https://doi.org/10.1016/j.enbuild.2024.114595.

[7] M. Liu, **M. Guo**, Y. Fu, Z. O'Neill, Y. Gao, Expert-guided imitation learning for energy management: Evaluating GAIL's performance in building control applications, Appl. Energy 372 (2024) 123753. https://doi.org/10.1016/j.apenergy.2024.123753.

[8] Z. Pang, **M. Guo**, B. Smith-Cortez, Z. O'Neill, Z. Yang, M. Liu, B. Dong, Quantification of HVAC energy savings through occupancy presence sensors in an apartment setting: Field testing and inverse modeling approach, Energy and Buildings 302 (2024) 113752. https://doi.org/10.1016/j.enbuild.2023.113752.

[9] D. Wang, M. Li, **M. Guo**, Q. Shi, C. Zheng, D. Li, S. Li, Z. Wang, Modelling variable refrigerant flow system for control purpose, Energy and Buildings 292 (2023) 113163. https://doi.org/10.1016/j.enbuild.2023.113163.

[10] H. Zhong, **M. Guo**, Y. Wang, Z. Wang, Quantify the magnitude and energy impact of overcooling in a sub-tropical campus building, Build. Environ. (2023) 110033.https://doi.org/10.1016/j.buildenv.2023.110033.

[11] Y. Chen, **M. Guo**, Z. Chen, Z. Chen, Y. Ji, Physical energy and data-driven models in building energy prediction: A review, Energy Rep. 8 (2022) 2656–2671. https://doi.org/10.1016/j.egyr.2022.01.162. [**Highly Cited Paper**]

[12] **M. Guo**, P. Xu, T. Xiao, R. He, M. Dai, S.L. Miller, Review and comparison of HVAC operation guidelines in different countries during the COVID-19 pandemic, Build. Environ. 187 (2021) 107368. https://doi.org/10.1016/j.buildenv.2020.107368. [**Highly Cited Paper**]

CONFERENCE PAPERS

[1] H.H. Ghoneim, **M. Guo**, Z. Yang, Z. O'Neill, Assessing Room-Level Thermal Variation in a Single-Family Home Through a Long-Term Thermostat Reference and Temperature Distribution Analysis, 2026 ASHRAE Winter Conference, Las Vegas, United States, Jan. 31-Feb. 4, 2026. https://doi.org/10.63044/w26gho24.

[2] L. Pinheiro, **M. Guo**, Z. Wang, Z. Pang, Z. O'Neill, X. Lu, Uncertainty Quantification of Occupancy Sensor Errors on Energy Savings and Thermal Comfort in Residential Homes, 2026 ASHRAE Winter Conference, Las Vegas, United States, Jan. 31-Feb. 4, 2026.

[3] **M. Guo**, Z. Pang, Z. O'Neill, B. Smith-Cortez, Z. Yang, B. Dong, Long-Term Field Testing of Occupant Counting Sensors and Occupancy-Centric HVAC Controls in an Office Building, 2025 ASHRAE Winter Conference, Orlando, United States, February 8-12, 2025.

[4] L. Pinheiro, **M. Guo**, Z. Pang, Z. O'Neill, Quantifying Energy Savings of Occupancy-Centric HVAC Controls Utilizing Longitudinal Occupancy Sensing Data and Calibrated Energy Simulations, 2025 ASHRAE Winter Conference, Orlando, United States, February 8-12, 2025.

[5] **M. Guo**, Y. Fu, M. Liu, Z. O'Neill, Investigations on the Influence of Model Accuracy in Deep Reinforcement Learning Control for HVAC Application, 2024 ASHRAE Winter Conference, Chicago, United States, January 20-24, 2024.

[6] **M. Guo**, P. Xu, H. Wang, J. Gu, Z. Chen, Building energy modelling based on building information modelling: the remaining problems and a more robust method, In Proceedings of the 18th International Building Simulation Conference of IBPSA, Shanghai, China, September 4-6, 2023: pp. 1022–1030. https://doi.org/10.26868/25222708.2023.1412

MANUSCRIPTS IN PREPARATION

[1] **M. Guo**, Z. Qi, S. Zhan, Q. Zhu, Z. O'Neill, Benchmarking Intelligent Controls in Building HVAC Systems: an Extensible and Modular Framework and a Case Study

[2] **M. Guo**, Z. O'Neill, Z. Yang, A Co-simulation Platform for Energy-Saving Evaluation of HVAC Control Strategies in Cleanrooms

[3] **M. Guo**, Y. Choi, Z. O'Neill, S.-M. Cheong, What Matters Most? Feature Importance of Demographics, Tree Canopy, and Temperature for ZIP Code–Level Energy Use Intensity

POSTERS AND INVITED PRESENTATION

[1] **M. Guo**, S.-M. Cheong, Z. O'Neill, Engineering and Policy Insight into Residential Electricity Demand Resilience in a Changing Climate: Evidence from Large-Scale Smart Meter Data, oral presentation in Texas A&M Conference on Energy, College Station, United States, September 25, 2025.

[2] **M. Guo**, Z. O'Neill, Z. Yang, C. Chang, An Integrated Approach to Modeling, Decision-making, and Control for Energy Efficient Manufacturing, poster presentation in 2025 Texas HSRUs and HSIs: STEAM Research showcase, College Station, United States, September 11, 2025.

[3] **M. Guo**, Z. O'Neill, S. Cheong, Characterizing Residential Electricity Consumption in Harris County: Key Patterns and Demographic Influences, poster presentation in 2024 Texas Energy Summit, Austin, United States, November 19-21, 2024.

[4] **M. Guo**, Z. Pang, Z. O'Neill, Prediction of Occupant Number and Identification of Occupant Behaviors via Learning Algorithms – A Case Study in a Small Office Building. Seminar in Proceedings of the 2023 ASHRAE Annual Conference, Tampa, U.S., June 24-28, 2023.

HONORS AND AWARDS

2025	J. Mike Walker '66 Impact Award in Mechanical Engineering, TAMU	
2025	Melbern G. Glasscock '59 Endowed Graduate Fellowship in Mechanical Engineering, TAMU	
2025	ASHRAE Graduate Grant-in-Aid Award	
2024	Brenda and Jerry Gray '62 Fellowship in Mechanical Engineering, TAMU	
2024-2025	Graduate Travel Award from Department of Mechanical Engineering	
2022	Outstanding Graduate of Shanghai Municipality	1%
2021	Outstanding Students, Tongji University	2%
2021	National Scholarship, Ministry of Education of the People's Republic of China	0.2%
2020	3rd prize of Yada Scholarship, Tongji University	
2019	Outstanding Graduate of Chongqing Municipality	1%
2017-2019	National Encouragement Scholarship three times, Chongqing University	5%
2018	Outstanding Students, Chongqing University	5%
2017	Excellent League Member, Chongqing University	5%

COMPETITION AWARDS

Global AI Competition March 2022

Organizer: the Electrical and Mechanical Services Department (EMSD) of the Government of Hong Kong Special Administrative Region and the Guangdong Provincial Association for Science and Technology

• AWARD: **Best Use of Tencent Cloud** award, **Huawei Most Innovative Use of Data** award and **Gold award** (Team leader, 1st place on the leaderboard)

• Description: Predict the hourly cooling load for the next three months using approximately one and a half years of historical data.

IFLYTEK A.I. Developer Competition: Demand Strategy Optimization Based on Flexible Load June to October 2022

Organizer: Sun Power

• AWARD: First Place (First author, First prize)

• Description: Rearrange the flexible load and battery schedule for 5 months in 5-minute intervals to reduce the electricity cost.

PROFESSIONAL SOCIETIES AND SERVICE

Reviewer 2023 to present

• Journal: Science and Technology for the Built Environment

• Conference: ASHRAE, IBPSA and CLIMA conference papers

• Volunteer for ASHRAE Handbook—HVAC Applications, Chapter 43

Membership and service

• Student member of ASHRAE, 2023 to present

• Officer (webmaster) of ASHRAE TAMU Student Branch, Fall 2023 - Spring 2025

TEACHING EXPERIENCE

Research Mentor *@Texas A&M University* 2024 to present

• Mentored undergraduate and early-stage Ph.D. students in research projects, resulting in co-authored conference publications.

• Mentored a high school student on a scientific project using ResStock to evaluate the energy impacts of rooftop photovoltaic installation.

Academy for Future Faculty Program *@Texas A&M University* 2025 - 2026

• Participated in a faculty development program through the Center for the Integration of Research, Teaching, and Learning (CIRTL), focusing on teaching and other faculty skills in STEM disciplines.

Teaching Assistant *@Tongji University* 2020 - 2021

Business Administration in the Master of Engineering Management (MEM) program

• Managed attendance records, coordinated group discussion schedules, and monitored the progress of course projects.

SKILLS

Programming	Python, C#, C, R (novice)
Protocols & APIs	gbXML, Revit SDK
Simulation	EnergyPlus, ResStock, Modelica (Dymola), Fluent (novice)
Modeling	AutoCAD, Revit, SketchUp, Grasshopper (novice)

Last Updated: Oct. 2026